IME 775 — Lecture 8

Probability Foundations for Machine Learning

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1. Why Probability in ML?

Machine learning models must handle **uncertainty**:

- Noisy sensor readings
- Incomplete data
- Inherent randomness in natural processes

Key idea: We model data as outcomes of random experiments, then learn patterns from observed outcomes.

ML Connection: Training = estimating probability distributions from data

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2. Probability Basics

Sample Space and Events

Sample space Ω: set of all possible outcomes

Event $E \subseteq \Omega$: a subset of outcomes

Examples:

Experiment	Sample Space Ω	Event E
Coin flip	$\{H, T\}$	$E = \{H\}$

Die roll	$\{1, 2, 3, 4, 5, 6\}$	$E = \{2, 4, 6\}$ (even)
Image class	\{$cat, dog, bird}$	E = \{$cat}$

Probability of an Event

$$P(E) = \frac{|E|}{|\Omega|} \quad \text{(equally likely outcomes)}$$

Properties:

- $0 \leq P(E) \leq 1$
- $P(\Omega) = 1$
- $P(\emptyset) = 0$
- $P(\bar{E}) = 1 - P(E)$

Workout: A bag has 3 red, 2 blue, 5 green balls. What is $P(\text{red or blue})$?

Solution: $P(\text{red or blue}) = \frac{3+2}{3+2+5} = \frac{5}{10} = 0.5$

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3. Random Variables

A **random variable** X is a function that maps outcomes to numbers:

$$X : \Omega \to \mathbb{R}$$

Discrete Random Variables

Takes countable values: $X \in \{x_1, x_2, \ldots\}$

Probability Mass Function (PMF):

$$p(X = x_i) = P(\{\omega \in \Omega : X(\omega) = x_i\})$$

Example: X = number on a fair die

$$p(X = k) = \frac{1}{6}, \quad k \in \{1, 2, 3, 4, 5, 6\}$$

Continuous Random Variables

Takes values in an interval: $X \in \mathbb{R}$

Probability Density Function (PDF): $f(x)$ such that

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx$$

Key property: $f(x) \geq 0$ but $f(x)$ can exceed 1! Only areas represent probabilities.

Cumulative Distribution Function (CDF)

$$F(x) = P(X \leq x)$$

For continuous: $F(x) = \int_{-\infty}^{x} f(t)\,dt$

For discrete: $F(x) = \sum_{x_i \leq x} p(X = x_i)$

Workout: If $f(x) = 2x$ for $x \in [0, 1]$ and 0 otherwise, find $P(0.5 \leq X \leq 1)$.

Solution:

$$P(0.5 \leq X \leq 1) = \int_{0.5}^{1} 2x\,dx = \left[x^2\right]_{0.5}^{1} = 1 - 0.25 = 0.75$$

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4. Joint and Marginal Probability

Joint Probability

For two random variables X and Y:

$$P(X = x_i, Y = y_j) = P(X = x_i \text{ and } Y = y_j)$$

Marginal Probability (Sum Rule)

$$P(X = x_i) = \sum_j P(X = x_i, Y = y_j)$$

Intuition: "Sum out" the variable you don't care about.

Example: Classify images by (animal, color):

	Brown	White	Black
Cat	0.10	0.15	0.05
Dog	0.20	0.10	0.10
Bird	0.05	0.10	0.15

$$P(\text{Cat}) = 0.10 + 0.15 + 0.05 = 0.30$$

$$P(\text{Brown}) = 0.10 + 0.20 + 0.05 = 0.35$$

Product Rule

$$P(X, Y) = P(X|Y) \cdot P(Y) = P(Y|X) \cdot P(X)$$

Independence

X and Y are independent iff:

$$P(X, Y) = P(X) \cdot P(Y)$$

Workout: From the table above, are "Cat" and "Brown" independent?

Solution:

- $P(\text{Cat, Brown}) = 0.10$
- $P(\text{Cat}) \cdot P(\text{Brown}) = 0.30 \times 0.35 = 0.105$
- Since $0.10 \neq 0.105$, they are **not independent**.

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5. Sampling from Distributions

Sampling = drawing values according to a distribution.

Given PMF $p(X = x_i)$, sampling produces values where each x_i appears with frequency $\approx p(X = x_i)$.

Why Sampling Matters in ML

- **Training data** is a sample from the true data distribution
- **Mini-batch SGD** samples subsets for gradient estimation
- **Monte Carlo methods** estimate integrals via sampling
- **Data augmentation** generates new samples

Empirical Distribution

Given N samples $\{x^{(1)}, \ldots, x^{(N)}\}$:

$$\hat{p}(X = x_i) = \frac{\text{count of } x_i}{N}$$

As $N \to \infty$, $\hat{p} \to p$ (Law of Large Numbers).

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6. Expected Value (Mean)

Discrete Case

$$E[X] = \sum_i x_i \cdot p(X = x_i)$$

Continuous Case

$$E[X] = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

Properties:

- $E[aX + b] = aE[X] + b$ (linearity)
- $E[X + Y] = E[X] + E[Y]$ (always, even if dependent)
- $E[XY] = E[X]E[Y]$ (only if independent)

Workout: A fair die: $E[X] = ?$

Solution:

$$E[X] = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{21}{6} = 3.5$$

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7. Variance and Standard Deviation

Variance

$$\mathrm{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$$

Standard Deviation

$$\sigma = \sqrt{\mathrm{Var}(X)}$$

Properties:

- $\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$
- $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ (if independent)

Workout: For $X \sim \mathrm{Uniform}\{1, 2, 3, 4, 5, 6\}$, compute $\mathrm{Var}(X)$.

Solution:

- $E[X] = 3.5$ (from above)
- $E[X^2] = \frac{1}{6}(1 + 4 + 9 + 16 + 25 + 36) = \frac{91}{6} \approx 15.17$
- $\mathrm{Var}(X) = E[X^2] - (E[X])^2 = \frac{91}{6} - \frac{49}{4} = \frac{182-147}{12} = \frac{35}{12} \approx 2.92$

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8. Multivariate Expected Value

For a random vector $\vec{X} = (X_1, X_2, \ldots, X_d)^T$:

$$E[\vec{X}] = \begin{pmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_d] \end{pmatrix} = \vec{\mu}$$

This is just the **mean vector** — same concept as in PCA!

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9. Covariance and Covariance Matrix

Covariance (2 variables)

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

- $\text{Cov}(X, Y) > 0$: X and Y increase together
- $\text{Cov}(X, Y) < 0$: one increases as other decreases
- $\text{Cov}(X, Y) = 0$: no linear relationship

Covariance Matrix

For $\vec{X} = (X_1, \ldots, X_d)^T$:

$$\Sigma = E[(\vec{X} - \vec{\mu})(\vec{X} - \vec{\mu})^T]$$

$$\Sigma_{ij} = \text{Cov}(X_i, X_j)$$

Properties:

- Σ is **symmetric**: $\Sigma_{ij} = \Sigma_{ji}$
- Σ is **positive semi-definite**: $\vec{v}^T \Sigma \vec{v} \geq 0$
- Diagonal entries = variances: $\Sigma_{ii} = \text{Var}(X_i)$

Connection to PCA: The covariance matrix Σ is exactly the matrix whose eigenvectors are the principal components!

Workout: Given $\Sigma = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}$, what is $\text{Var}(X_1)$, $\text{Var}(X_2)$, and $\text{Cov}(X_1, X_2)$?

Solution:

- $\text{Var}(X_1) = \Sigma_{11} = 4$
- $\text{Var}(X_2) = \Sigma_{22} = 3$
- $\text{Cov}(X_1, X_2) = \Sigma_{12} = 2$ (positive → variables increase together)

Key Takeaways

1. **Random variables** map outcomes to numbers; described by PMF (discrete) or PDF (continuous)
2. **Joint probability** captures relationships; **marginal** is obtained by summing out variables
3. **Expected value** is the "average" outcome; **variance** measures spread
4. **Covariance matrix** captures all pairwise relationships and connects directly to PCA
5. These concepts are the language of machine learning — losses, likelihoods, and Bayesian methods all build on them

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PyTorch Connection

```
import torch

# Sampling from distributions
uniform = torch.distributions.Uniform(0, 1)
samples = uniform.sample((1000,))

# Mean and variance
print(f"Mean: {samples.mean():.4f}")     # ≈ 0.5
print(f"Var:  {samples.var():.4f}")       # ≈ 1/12 ≈ 0.0833

# Covariance matrix from data
data = torch.randn(100, 3)  # 100 samples, 3 features
cov_matrix = torch.cov(data.T)
print(f"Covariance matrix:\n{cov_matrix}")
```